What changed for the double well?

Experiment version: `d20b401` Prepared: 8 October 2026

The main change was to add two drift estimates from observed increments, alongside the existing Method 1, Method 2, and ODE2. One new fit allows general polynomials; the other supplies the known odd symmetry of the drift. **The improved result with odd symmetry is a separate comparison, not an improvement to the original drift equations.**

1. Process and simulation

The supplied image was interpreted as the Itô SDE

$$dX_t = \underbrace{(0.5X_t - X_t^3)}_{\mu(X_t)}\, dt + \underbrace{(1 + 0.25X_t^2)}_{\sigma(X_t)}\, dB_t^P.$$

The first differential on the right of the image was treated as a typo for dt. The drift is $-V'(x)$ for $V(x) = x^4/4 - x^2/4$, with wells at $x = \pm 1/\sqrt{2}$.

Setting	Previous OU experiment	Double well
Stored samples N	100,000	1,000,000
Observation horizon T	1	200
Fine time step $\Delta t = T/N$	10^{-5}	2×10^{-4}
Initial state X_0	1	1
Random seed	3	42
Signature window M	100	100
Coarse samples $K = N/M$	1,000	10,000
Coarse time step $h = M\Delta t$	0.001	0.02

The longer horizon lets the path visit both wells. The fine step is also larger than in OU; this is not merely the same discretization run for longer. The existing Euler simulation is reused. State, time, and Brownian samples are aligned as in OU. There are $N - 1$ observed increments, with last recorded time $T - \Delta t$.

The recorded state range is $[-2.57273, 2.76237]$; the path is below zero for approximately 57.18% of the recorded samples. The seed was fixed, not selected by comparing errors across seeds.

2. Files added and code reused

- `data_generation/double_well.py`: process coefficients and simulation settings.
- `models/increment_drift.py`: the two new increment-based comparisons.
- `experiments/double_well/`: a separate executed notebook and figures.

Existing diffusion, signature, drift-recovery, noise-recovery, and SSISDE model files were reused without modification. Earlier experiments were unchanged. The README was extended to describe the new case.

3. Diffusion regression: same method, different search grid

The target is a degree-4 polynomial for $q(x) = \sigma^2(x)$. The complete library remains $\{1, x, x^2, x^3, x^4\}$; even symmetry is *not* imposed here. The exact variance, used only for comparison, is

$$q(x) = (1 + 0.25x^2)^2 = 1 + 0.5x^2 + 0.0625x^4.$$

As in OU, the code computes cumulative quadratic variation from every preceding squared increment, differentiates it with the existing order-3 finite-difference method, and fits SR3. Derivatives of $\hat{\sigma} = \sqrt{\hat{q}}$ come analytically from the fitted polynomial.

Setting	OU	Double well
SR3 thresholds	0.1, 0.3, 1, 3	0.1, 1, 3, 10
Relaxation parameter ν	0.1, 0.5, 2	Same
Iterations / tolerance	10,000 / 10^{-8}	Same
Selection	Last 20% of the path	Same

The fit uses the first 80% for training, leaving a three-sample gap before validation. Selection rejects nonconverged fits or nonpositive predicted variance on the observed path. The selected settings are refitted on all samples. The true diffusion is not used for selection.

The selected threshold is 1, with $\nu = 0.5$. The full-data estimate is

$$\hat{q}(x) = 1.00198797 + 0.50032259x^2 + 0.05995041x^4, \qquad \mathrm{RMSE}(\hat{\sigma}) = 0.00170021.$$

This RMSE is for $\sqrt{\hat{q}}$, not for $\hat{q}$. Positivity on observed states does not, in general, guarantee positivity outside their range.

4. Original drift methods: unchanged, with a new diagnostic

The signature stage still uses $M = 100$, ridge parameter 0.01, and the same six signature columns. Methods 1 and 2 receive $\mu(X_0) = -0.5$. ODE2 also receives the *known spatial slope* $\mu'(X_0) = -2.5$. No estimated-slope ODE2 experiment was added.

For Method 2, the implemented expression gives $b = 0$. Its continuous state-space equation $\mu' = (\sigma'/\sigma)\mu$ therefore preserves μ/σ. The double-well ratio $(0.5x - x^3)/(1 + 0.25x^2)$ is not constant. The numerical recurrence can introduce additional error along the fluctuating path.

ODE2 requires

$$\mu'' = A\mu' + B\mu, \qquad A = \frac{2\sigma'}{\sigma}, \qquad B = \frac{\sigma''}{\sigma} + \frac{2(\sigma')^2}{\sigma^2}.$$

At $x = 1$, the exact functions give $A = 0.8$, $B = 0.72$, and

$$\mu'' - A\mu' - B\mu = -6 - 0.8(-2.5) - 0.72(-0.5) = -3.64 \neq 0.$$

Thus, even exact diffusion cannot make this ODE2 equation exact for the stated drift. This diagnostic is not fed into the estimators. Method 1 also performs poorly in this run; that empirical result is distinct from the algebraic checks above.

5. New comparison: drift from observed increments

The additional fit uses only the fine-grid states and Δt. For a polynomial library $\phi(x)$, it estimates

$$\Delta X_i \approx \phi(X_i) c \, \Delta t.$$

For each training fold, let s_j be the root-mean-square value of library column j, and let $Z_{ij} = \phi_j(X_i)/s_j$. The fitted scaled coefficients solve

$$\hat{\beta} = \arg\min_{\beta} \left\{ \frac{1}{\Delta t} \|\Delta X - \Delta t Z \beta\|_2^2 + \lambda \|\beta\|_2^2 \right\}, \qquad \hat{c}_j = \hat{\beta}_j / s_j.$$

This corresponds to the implemented system $(\Delta t \, Z^\top Z + \lambda I)\hat{\beta} = Z^\top \Delta X$. Column scales are estimated from each training fold, not its validation data.

The search uses degrees $1, 2, 3, 4, 5$, penalties $\lambda \in \{0, 0.01, 0.1, 1, 10\}$, and five expanding time-series folds. The validation score is

$$L = \frac{1}{|\mathcal{V}|} \sum_{i \in \mathcal{V}} \left(\hat{\mu}(X_i)^2 - \frac{2\hat{\mu}(X_i)\Delta X_i}{\Delta t} \right).$$

This is squared error against $\Delta X_i / \Delta t$ after subtracting a candidate-independent term. Negative scores are allowed. The code finds the lowest mean score, admits models within one standard error of that score, chooses the smallest degree, and then the lowest mean score at that degree. It refits on the full path.

The selected general fit has degree 3 and $\lambda = 1$:

$$\hat{\mu}_{\text{general}}(x) = -0.0967098 + 0.763423x - 0.0536947x^2 - 1.18526x^3.$$

It supplies neither the true initial drift nor the initial slope. It is a separate long-observation comparison, not a replacement for signature-based estimation of ψ_{21}.

6. New comparison with supplied odd symmetry

The second fit supplies the property $\mu(-x) = -\mu(x)$. Its candidate libraries are

$$\{x\}, \qquad \{x, x^3\}, \qquad \{x, x^3, x^5\}.$$

It uses the same penalties and five-fold selection rule. The selected degree is again 3, with $\lambda = 1$:

$$\boxed{\hat{\mu}_{\text{odd}}(x) = 0.782957x - 1.17707x^3.}$$

The true coefficients 0.5 and -1 are not supplied. Odd symmetry, including $\mu(0) = 0$, *is* supplied from the stated model; it was not discovered from data. Neither increment fit is constrained to match $\mu(1) = -0.5$ or $\mu'(1) = -2.5$.

The true Brownian path is not an input to either fit. The degree, penalty, and SR3 parameters are selected from observed data, but the additional symmetry itself is prior information. Reported errors are measured after refitting on the same trajectory, not on an independent test path.

7. Noise recovery and final SSISDE fit

All five comparisons use the same estimated diffusion, recovered B^Q, and coarse step $h = 0.02$. Noise recovery is unchanged:

$$\Delta\hat{B}_i^P = \Delta\hat{B}_i^Q - \frac{\hat{\mu}(X_i)}{\hat{\sigma}(X_i)}h, \qquad \hat{B}_0^P = 0.$$

Small drift biases can accumulate when these increments are summed. A lower pointwise drift RMSE does not necessarily imply a lower Brownian-path RMSE.

Drift estimate	Drift RMSE	B^P path RMSE	ΔB^P RMSE
Method 1	1.347405	5.230128	0.017500
Method 2	1.197744	6.079452	0.013957
ODE2, known slope	1.201497	4.505206	0.014134
General increment fit	0.201668	12.868740	0.004806
Odd increment fit	0.149874	0.390244	0.004246

These are coarse-grid errors before SSISDE. Each RMSE is the square root of the mean squared difference from the corresponding simulated drift, Brownian path, or Brownian increments.

The final SSISDE implementation is unchanged and retains degree-5 libraries for both drift and diffusion. Its search settings differ from OU:

Setting	OU	Double well
Time-series folds	7	5
Elastic-net α grid	35 values, 10^{-5} to 10	21 values, 10^{-6} to 1
Mixing values ρ	$.3, .4, .5, .7, .85, .95, 1$	$.5, .9, 1$
Selection rule	One standard error	Same

The smaller search is used for the longer coarse trajectory. The odd-noise comparison uses the *general* degree-5 SSISDE libraries, not an odd-only final library.

Noise supplied to SSISDE	Final drift RMSE	Final diffusion RMSE
Method 1	1.200266	0.022644
Method 2	1.200266	0.018175
ODE2, known slope	1.200266	0.018040
General increment fit	0.197696	0.020209
Odd increment fit	0.146471	0.020118

For the odd-noise comparison, the selected final model is

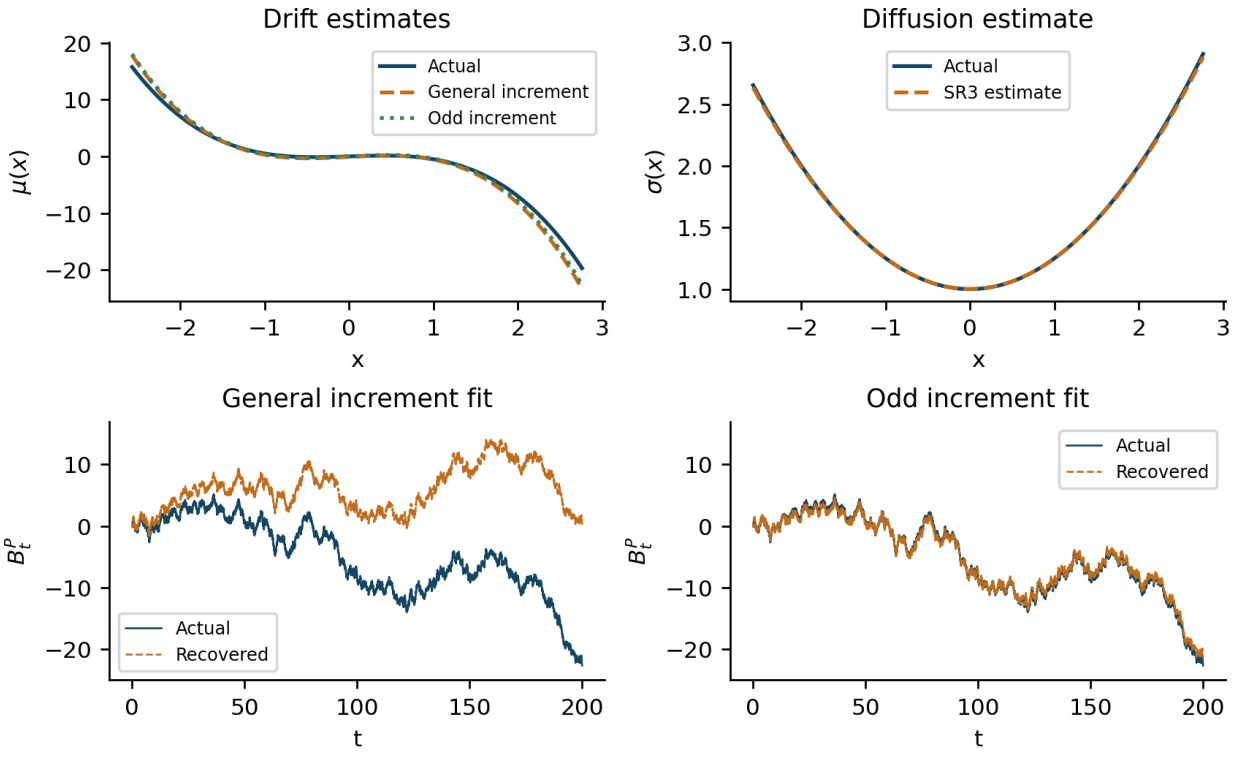
$$\hat{\mu}_{\text{SSISDE}}(x) = 0.815064x - 1.253827x^3 + 0.0202219x^5,$$
$$\hat{\sigma}_{\text{SSISDE}}(x) = 1.011768 + 0.224561x^2.$$

Its $\alpha = 0.00199526$ and $\rho = 0.5$. The final diffusion is less accurate than the earlier SR3 estimate. SSISDE selects zero drift for the first three noise estimates. No convergence warnings were recorded for these final fits.

8. What the results show

Diffusion is estimated accurately on the observed states. The original three drift methods do not recover the cubic drift accurately in this run. The additional odd-symmetry fit improves drift and Brownian-path recovery, but its coefficients still differ from the exact $0.5x - x^3$.

The general increment fit has smaller drift error than the original methods but larger accumulated noise error. All comparisons are retained. Small SSISDE increment error is not independent evidence of drift recovery because the reconstructed noise already depends on the estimated drift.



Comparison from the saved run. The upper panels show coefficient estimates before SSISDE; the lower panels compare the true Brownian path with the two increment-based reconstructions. Odd symmetry is supplied only in the labelled comparison.

9. Reproduce and inspect

From the repository root, run the notebook with the existing environment:

```
.venv/bin/python scripts/run_experiment.py
experiments/double_well/Stochastic_SINDy_Double_Well.ipynb
-in-place
```

Join the three lines as one command, or open the notebook and choose **Run All**. It saves numerical results in `outputs/double_well/results.json`, tuning tables in the same folder, and figures under `experiments/double_well/`. The notebook retains its displayed outputs.

Scope. This note describes commit `d20b401` without changing the experiments. Results come from its saved JSON and path arrays for one trajectory, not an independent test or a multi-seed study.